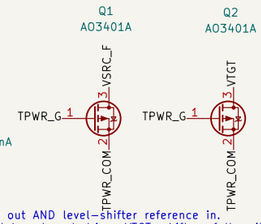


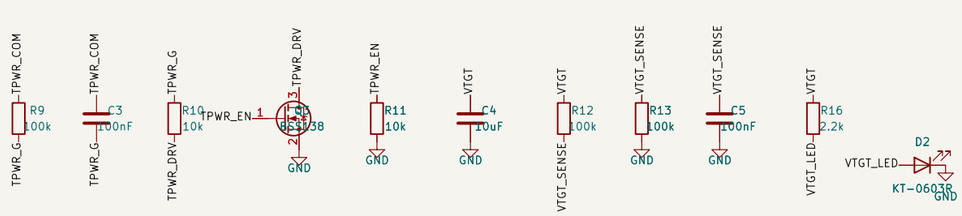


Channels (U2 A -> B, JPn bypass, Rn 33R -> connectors)
IO0: SWCLK / RVSWD CLK / SPI SCK / TCK / ISP SCK / LA0
IO1: SWDIO / SWIO / RVSWD DIO / UPDI / SPI MOSI / TMS / ISP MOSI / LA1
IO2: UART TX / SPI MISO / ISP MISO / LA2
IO3: UART RX / SPI CS / SWO alt / LA3
IO4: nRESET / UART CTS / ESP32 EN / nSRST / ISP RESET / LA4
IO5: SWO / UART RTS / ESP32 IO0 / TDO / LA5
IO6: TDI / LA6
IO7: LA7

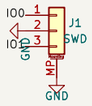
RP2040 pins
GP2–GP9 = IO0–IO7 A side (contiguous for PIO / logic analyser)
GP2/3/4/5 = SPI0 SCK/TX/RX/CSn, GP4/5/6/7 = UART1 TX/RX/CTS/RTS
GP10 = SHIFT_OE (U2 OE, R14 pull-down; raise only once VTGT (ADC0 x 2) >= 3.1 V, i.e. ADC0 >= 1.55 V)
GP14 = TPWR_EN (target power), GP26/ADC0 = VTGT_SENSE (VTGT/2)
GP16 = WS2812 on module, BOOT button = mode
Free: GP0, GP1, GP11, GP12, GP13, GP15, GP27, GP28, GP29

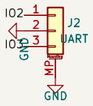


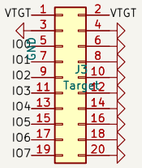


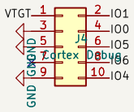


Target power: VTGT = target supply out AND level-shifter reference in.
Self-powered target: leave TPWR_EN low, target drives VTGT, shifters follow it.
Probe-powered target: firmware checks VTGT_SENSE ~0 V first, then sets TPWR_EN.
Q1+Q2 back-to-back: no back-feed into 3V3/5V when off.











Mode / channel map (firmware)
1 SWD+UART (green): IO0 SWCLK, IO1 SWDIO, IO2 TX, IO3 RX, IO4 nRESET, IO5 SWO
2 WCH (purple): CH32V003 SWIO on IO1; CH32V2xx/V3xx/X0 RVSWD IO1 data + IO0 clock; IO2 TX, IO3 RX; TPWR power cycle for unbrick
3 UPDI (yellow): IO1 UPDI, IO2 TX, IO3 RX
4 UART bridge (white): IO2 TX, IO3 RX, IO4 CTS, IO5 RTS; ESP32 auto-reset (esptool): host RTS -> IO4 (EN), host DTR -> IO5 (IO0/BOOT)
5 Logic analyser (blue): IO0–IO7 = CH0–CH7
6 SPI flash / serprog (orange): IO0 SCK, IO1 MOSI, IO2 MISO, IO3 CS
7 JTAG (cyan): IO0 TCK, IO1 TMS, IO5 TDO, IO6 TDI, IO4 nSRST
8 AVR ISP (red): IO0 SCK, IO1 MOSI, IO2 MISO, IO4 RESET

Open hardware; RP2040-Zero footprint from harshitgoel96/RP2040-Zero-Kicad-Fixed
Switched target power Q1/Q2, 3V3 or 5V by JP9, VTGT on ADC0
IO0–IO7 via optional TXS0108E (VTGT 3.3–5.5 V), JP1–8 bypass
SWD+UART, WCH SWIO/RVSWD, UPDI, UART, logic analyser, SPI flash, JTAG, AVR ISP

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Title: tinyLINK multi-mode debug probe (RP2040-Zero carrier)

Size: A3	Date: 2026-10-28	Rev: A
KiCad E.D.A. 10.0.3		Id: 1/1